- **Build my own lab**
  - A small but impact team
  - Recruit & train students
- **Keep a research record**
  - Papers on top conferences
  - Organize AI seminars
- **Academic services**
  - AC on AI/ML conferences
  - IEEE / ACM

Research Team & Goals

- **High-quality teaching**
  - Undergraduate courses
  - Graduate courses
- **Develop new courses**
  - AI on engineering problems
  - Cutting-edge technologies
- **Scientific activities**
  - Interdisciplinary seminars
  - Science Outreach

Teaching

- **Year 1-2 (Plan 1)**
  - **NSF Core (CISE / IIS)**
  - **ONR & AFOSR**
  - **Industry grants**

- **Year 3-5 (Plan 1 & 2)**
  - **NSF CAREER Award**
  - **DARPA & NASA**
  - **Industry grants**

Funding! Funding!!

Expected Outcomes (3–5 Years)

YOU GET FUNDING!
YOU GET FUNDING!
gifmemes.io

YOU GET FUNDING!
YOU GET FUNDING!
gifmemes.io

Expected Outcomes (3–5 Years)

Teaching

- **High-quality teaching**
  - Undergraduate courses
  - Graduate courses
- **Develop new courses**
  - AI on engineering problems
  - Cutting-edge technologies
- **Scientific activities**
  - Interdisciplinary seminars
  - Science Outreach

Research Team & Goals

- **Build my own lab**
  - A small but impact team
  - Recruit & train students
- **Keep a research record**
  - Papers on top conferences
  - Organize AI seminars
- **Academic services**
  - AC on AI/ML conferences
  - IEEE / ACM

Funding! Funding!!

- **Year 1-2 (Plan 1)**
  - **NSF Core (CISE / IIS)**
  - **ONR & AFOSR**
  - **Industry grants**
- **Year 3-5 (Plan 1 & 2)**
  - **NSF CAREER Award**
  - **DARPA & NASA**
  - **Industry grants**

References

4 AI for Real-world Applications

- **FairFedMed:** Group Fairness in Federated Medical Imaging (TMI 2025)

3 Controllable Generation & Image and *Video* Editing

- **DVRF**: Delta velocity rectified flow for text-to-image editing (CVPR 2026)
- **DNAEdit:** Text-Guided Rectified Flow Editing (NIPS 2025 Spotlight) ★
- **SplitFlow:** Inversion-Free Text-to-Image Editing (NIPS 2025)
- **FiVE-Bench:** Video Editing Benchmark for Diffusion Models (ICCV 2025) ★
- **Exact Feature Matching:** Style Transfer & Domain Gen (CVPR 2022 Oral)

2 Consistent Understanding & *Video* Segmentation

- **VideoITG**: Multimodal Video Understanding (CVPR 2026)
- **MaSS13K:** Matting-level Semantic Segmentation Benchmark (CVPR 2025)
- **Spatial-Mamba:** Structure-aware State Fusion (ICLR 2025) ★
- **UniVS:** Unified Video Segmentation with Prompts (CVPR 2024) ★
- **MDQE:** Segmenting Occluded Instances in Video (CVPR 2023)
- **STMask**: Spatial calibration and temporal fusion for video segmentation (CVPR 2021)
- **One-to-Few:** Label Assignment for Dense Detection (CVPR 2023)

1 Robust Perception & *Video* Deraining

- **OT-MSCSC**: Online Rain/Snow Removal from Surveillance Videos (TIP 2021) ★
- **MS-CSC**: Video Rain Removal via Multiscale Convolutional Sparse Coding (CVPR 2018) ★